DEVELOPER ROADMAP

Stop Paying for GitHub Copilot: The 5-Step Local AI Setup

A complete step-by-step setup guide for a private, offline developer loop.

SLIDE 2

Step 1: Sizing to Your Workstation

The VRAM Bottleneck If parameters exceed GPU VRAM, the system swaps to RAM. Generation drops to **< 2 tokens/sec**. Keep models strictly inside VRAM.

Rule of Thumb: Size model footprint to fit within 70% of total VRAM, leaving 30% for OS/browser.

Hardware Compatibility

- **8 GB VRAM:** < 3B models (e.g. Qwen 1.5B Base)
- **16-24 GB VRAM:** 12B-14B models (e.g. Qwen 14B / Gemma 12B)
- **32 GB+ VRAM:** 22B+ models (e.g. Codestral 22B)

SLIDE 3

Step 2: Install Engine & Pull Weights

```
# 1. Download & Install Ollama from ollama.com

  # 2. Pull base model for tab autocomplete
$ ollama pull qwen2.5-coder:1.5b-base

  # 3. Pull instruct model for chat reasoning
$ ollama pull qwen2.5-coder:14b-instruct
```

Ollama schedules GPU memory and runs the local API server on port 11434.

Step 3: IDE Split-Model Configuration

1. config.json Configuration

```
{
  "models": [{"provider": "ollama",
"model": "qwen2.5-coder:14b"}],
  "tabAutocompleteModel": {
    "provider": "ollama",
    "model": "qwen2.5-coder:1.5b-
base"
  }
}
```

Replacing Copilot Features

- **Ghost Text Autocomplete:** Triggers automatically as you type. Press `Tab` to accept.
- **In-place Edit (`Cmd+I`):** Select code, press `Cmd+I` to refactor selection inline.
- **Sidebar Chat (`Cmd+L`):** Ask questions, reference files with `@file` context.

Step 4: Ignore Indexes & Solve Latency

1. .continueignore Exclusions
Prevent 100% CPU loops by excluding dependencies from embeddings index:

```
node_modules/
dist/
build/
.git/
```

2. Latency Prefix Tuning To solve autocomplete freeze when switching tabs, set:

- **Prefix Length:** `500`
- **Suffix Length:** `250`

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Step 5: Terminal CLI Agents & Pipes

```
# 1. Launch terminal coding agents directly via Ollama
$ ollama launch opencode      # Autonomous developer TUI
$ ollama launch copilot-cli   # Command generator helper
$ ollama launch claude        # Agentic coding CLI tool

  # 2. Or pipe log dumps directly for troubleshooting
$ cat error.log | \
    ollama run qwen2.5-coder:14b "Explain this error"
```

Ollama's desktop app includes a native registry to launch open-source terminal agents.

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Quick-Reference Architecture Map

The Core Workflow

- **Tab Autocomplete:** Route to `Qwen-1.5B-Base` (runs at **188.4 tok/s**).
- **Chat Reasoning:** Route to `Qwen-14B-Instruct` (runs at **30.0 tok/s**).
- **CLI Pipe:** `cat error.log | ollama run qwen2.5-coder:14b`

Hardware Sizing & CPU Fixes

- **VRAM Rule:** Keep models within VRAM bounds to prevent swap lags.
- **VS Code Settings:** Set autocomplete Prefix to `500` & Suffix to `250`.
- **CPU Protection:** Prevent CPU loops by adding compiled folders to `.continueignore`.

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Reclaim Your Code Privacy

Full configurations, custom `Modelfile` setups, and terminal pipes are in the full guide.

Connect & Build:

- **Substack:** praveenbuilds.substack.com
- **GitHub (Companion Code):** github.com/praveenveera/software-permanence
- **Dev.to:** dev.to/praveen_builds
- **Medium:** medium.com/@praveenveera92
- **Instagram:** @praveen.builds
- **LinkedIn:** linkedin.com/in/praveen-veera-6ab22567
- **Hashnode:** hashnode.com/@praveen-builds